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MANAGEMENT & RESEARCH(MBA & MCA),
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2025-2026

A
Project Report on

“Book My Parcel”

Using Backend- **NodeJs** Frontend- **React**

Submitted In Partial Fulfilment of The Requirement For

The Award of The Degree of

MCA Sem-IV

PROJECT GUIDE

Prof. Suraj Hingane

SUBMITTED BY

Priti Pralhad Tole

Project Guide

HOD

Internal Examine

External Examiner

Completion Certificate By Company (Company Letter Head)



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CERTIFICATE

This is to certify that **Priti Pralhad Tole** is a bonafide student of **Abhinav Education Society Institute Of Management & Research Pune**, has successfully completed the project work as prescribed by the Savitribai Phule Pune University, Pune, in the partial fulfilment of the **(MCA) Program 2025-2026**.

The project work, titled as “**Book My Parcel**” has been carried out at the establishment of “**Nimbja Security Solutions Pvt Ltd**”.

Project Guide

HOD

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DECLARATION

To HOD,

Abhinav Education Society's Institute Of Management & Research Pune

Respected Sir,

I, **Priti Pralhad Tole** hereby declare that the project entitled “**Book My Parcel**” developed and submitted under the guidance of “**Prof. Suraj Hingane** ” is my original work.

The system presented here is my own work and has not been duplicated from any other sources.

Thank you

Yours Sincerely

Priti Pralhad Tole

ACKNOWLEDGEMENT

I would like to express my gratitude and sincere regards to the following people to whom I am grateful for their support and help without which I would not have been able to do this project entitled “**Book My Parcel**”.

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Special thanks to my family for their constant support and encouragement. Your belief in me has been a great source of motivation.

Finally, I would like to acknowledge the authors and researchers whose work I have referenced. Their contributions have been essential to my project.

Thank you all for your support and assistance.

Student name

Priti Pralhad Tole

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CHAPTER 1: INTRODUCTION

Book My Parcel is a community-driven, technology-powered parcel delivery platform built to solve the growing challenge of affordable and accessible intercity parcel delivery in India. By connecting everyday travellers with individuals who need to send parcels, the platform creates a shared economy model that benefits all stakeholders — senders, travellers, and the broader community.

1.1 Company Profile

Book My Parcel is a technology startup concept developed as an academic project, representing a viable product-market fit for the Indian logistics and delivery space. The platform operates as a two-sided digital marketplace with the following organizational focus:

- **Mission:** To make intercity parcel delivery affordable, accessible, and community-driven by leveraging the everyday travel of millions of Indians.
- **Vision:** To become India's leading peer-to-peer parcel delivery network covering Tier-1, Tier-2, and Tier-3 cities.
- **Domain:** Logistics Technology / Collaborative Economy
- **Technology Stack:** Full-stack web application built with React.js, Node.js, PostgreSQL, and various third-party integrations.
- **Target Market:** Individual senders, small business owners, and regular intercity travellers across India.

The platform was designed with scalability, security, and ease-of-use in mind, aiming to provide a professional-grade delivery experience at a fraction of traditional courier costs.

1.2 Existing System

The current parcel delivery landscape in India relies heavily on traditional courier companies and informal arrangements:

- **Traditional Courier Services:** Companies like DTDC, BlueDart, and India Post dominate the market but are often expensive, slow, and limited in reach for Tier-2 and Tier-3 cities.
- **E-Commerce Logistics Arms:** Platforms like Amazon Logistics and Delhivery cater primarily to B2C e-commerce shipments, leaving individual senders with fewer options.
- **Informal Arrangements:** Many people rely on asking friends or relatives to carry parcels, which lacks transparency, accountability, and tracking.
- **Ride-Sharing Apps:** Platforms like Rapido and Ola do not provide structured intercity parcel delivery with verification and payment systems.

Key shortcomings of the existing system include:

- High delivery costs imposed by traditional courier companies.
- Slow delivery timelines, especially for intercity or semi-urban deliveries.
- No existing digital platform that connects everyday travellers with parcel senders.
- Limited reach of courier networks to smaller towns and villages.
- Lack of real-time tracking and transparent communication between sender and deliverer.

1.3 Scope of Work

The Book My Parcel platform covers all aspects of the intercity parcel delivery lifecycle, from parcel creation and traveller matching through payment processing and final delivery verification.

System Boundaries

The current scope includes a responsive web application supporting desktop and mobile browsers, targeting users across India. The platform handles end-to-end delivery workflows including user registration, route management, intelligent matching, secure OTP-based handover, digital payments, and admin oversight.

Target Users

- Individual senders — students, professionals, families needing to send parcels between cities.
- Small business owners — businesses sending samples, documents, or small goods.
- Regular intercity travellers — daily commuters, professionals, and students travelling between cities who want to earn extra income.
- Platform administrators — responsible for KYC approval, user management, and platform monitoring.

Geographical Scope

The platform is designed for use across India, with address validation and geocoding configured specifically for Indian addresses using the Google Address Validation API (India ML model). The system supports deliveries between any two cities or towns covered by Google Maps.

Out of Scope

- International parcel delivery.
- Native mobile application (iOS/Android) — planned as future enhancement.
- Heavy or commercial freight delivery.
- Automated AI-based customer support.

1.4 Operating Environment – Hardware and Software

Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3 / AMD Ryzen 3	Intel Core i5 / AMD Ryzen 5
RAM	4 GB	8 GB or more
Storage	10 GB Free Space	50 GB SSD
Network	Broadband Internet (2 Mbps)	High-Speed Internet (10 Mbps+)
Display	1280 x 720 resolution	1920 x 1080 (Full HD)

Software Requirements:

Category	Technology / Tool	Version / Details
Runtime Environment	Node.js	v18.x LTS or higher
Frontend Framework	React.js	v18.x
State Management	Redux Toolkit	v1.9.x
Backend Framework	Express.js	v4.x
Database	PostgreSQL with PostGIS	v15.x + PostGIS 3.x
ORM	Sequelize	v6.x
Real-time Communication	Socket.io	v4.x
Payment Gateway	Razorpay SDK	Latest
SMS Service	Twilio	Latest
Push Notifications	Firebase Cloud Messaging	Latest
Maps / Geocoding	Google Maps Platform APIs	Latest
Build Tool	Vite	v5.x
Version Control	Git / GitHub	Latest
Deployment – Frontend	Vercel	Cloud Platform
Deployment – Backend	Render	Cloud Platform
Operating System	Windows 10 / Ubuntu 20.04+	Any modern OS
Browser	Chrome, Firefox, Edge	Latest versions

CHAPTER 2: PROPOSED SYSTEM

2.1 Proposed System

Book My Parcel proposes a full-stack web application that operates on a two-sided marketplace model. The proposed system addresses all identified shortcomings of the existing delivery ecosystem through the following key innovations:

- **Intelligent Matching Engine:** A 5-stage geographic and temporal matching algorithm that pairs parcel senders with suitable travellers based on route, timing, capacity, and detour analysis.
- **OTP-Based Handover Verification:** Secure 4-digit OTP verification at both pickup and delivery stages ensures chain of custody and accountability.
- **Integrated Digital Payments:** Razorpay payment gateway supports upfront payment and post-delivery cash settlement, making the platform accessible to users at all levels of digital adoption.
- **Real-Time Communication:** Socket.io and Firebase Cloud Messaging ensure all parties receive live updates on booking status and parcel movement.
- **KYC Verification:** Admin-managed identity verification of all travellers builds platform trust and safety.

System Architecture

The system follows a three-tier client-server architecture. The Presentation Layer (React.js frontend) communicates with the Application Layer (Node.js/Express.js backend) through RESTful APIs and Socket.io connections. The Data Layer consists of a PostgreSQL database with PostGIS extensions for geospatial data management.

- **Frontend:** React.js deployed on Vercel — handles all user interactions, state management via Redux Toolkit, and real-time Socket.io connections.
- **Backend:** Node.js / Express.js deployed on Render — processes API requests, runs the matching engine, manages bookings, handles payments, and coordinates real-time events.
- **Database:** PostgreSQL with PostGIS — stores all user, parcel, route, booking, and payment records with spatial query capability.
- **External Services:** Google Maps Platform, Razorpay, Twilio SMS, and Firebase Cloud Messaging.

2.2 Objective of System

The key objectives of the Book My Parcel system are:

- To design and develop a full-stack web application that enables parcel senders to publish delivery requests and find matching travellers.
- To implement a 5-stage geographic and temporal matching engine that accurately pairs parcel delivery requests with suitable traveller routes.
- To integrate Google Maps Platform APIs for address validation, geocoding, route calculation, and intermediate city extraction.
- To implement a secure, OTP-based parcel handover verification system for both pickup and delivery stages.
- To integrate Razorpay payment gateway supporting both upfront and post-delivery payment modes.

- To develop a real-time notification system using Socket.io and Firebase Cloud Messaging.
- To build a multi-role authentication system supporting Sender, Traveller, and Admin roles.
- To implement a complete KYC (Know Your Customer) verification workflow managed by the Admin panel.
- To create an intuitive and responsive user interface using React.js and Redux for state management.
- To deploy the application across multiple environments using Render and Vercel.

2.3 User Requirements

Sender Requirements

- Register and log in with email and password.
- Create a parcel delivery request with pickup and delivery addresses using Google Places Autocomplete.
- View a list of matched travellers who have accepted the request.
- Select a preferred traveller and confirm booking with payment.
- Track booking status in real time (Confirmed → In Transit → Delivered).
- Receive OTP for parcel pickup verification and share with traveller.
- View booking history and delivery records.

Traveller Requirements

- Register with KYC document submission (Aadhaar, PAN, Selfie).
- Create one-time or recurring travel routes with transport mode and schedule.
- View and accept incoming parcel requests matching their route.
- Manage active bookings and execute OTP-based pickup and delivery flows.
- View earnings dashboard with per-booking payment records.

Admin Requirements

- Access admin dashboard with platform-level statistics.
- Review, approve, or reject traveller KYC documents.
- Manage users, bookings, routes, and parcel requests.
- Monitor all active and completed bookings on the platform.

2.4 Feasibility Study

Technical Feasibility

The system is technically feasible. All technologies used — React.js, Node.js, PostgreSQL, PostGIS, Socket.io, Razorpay, Google Maps, Twilio, and Firebase — are mature, well-documented, and widely deployed in production environments. The development team has demonstrated implementation capability through the working prototype.

Economic Feasibility

The platform operates on a commission-based revenue model. Operational costs include cloud hosting (Vercel and Render), database hosting, and third-party API usage fees. With a growing user base, these costs remain proportional to transaction volume, making the model economically viable. Initial investment is low given the use of cloud platforms and free-tier services during development.

Operational Feasibility

The system is operationally feasible. The two-sided marketplace model aligns with existing user behaviour — senders already look for affordable delivery options, and travellers already commute intercity. The OTP-based verification and admin KYC workflow ensure operational trust and accountability without requiring physical infrastructure.

Schedule Feasibility

The project was completed within the academic year timeline. Key milestones — requirements gathering, architecture design, implementation, testing, and documentation — were achieved in sequence. The modular architecture allowed parallel development of frontend and backend components.

2.5 Module Specifications

User (Sender) Module

The User Module provides a complete parcel management experience from creation to delivery confirmation.

- User Registration and Login with email/password authentication.
- Profile management with photo upload and personal details.
- 3-Step Parcel Creation Wizard: parcel details → traveller selection → payment.
- Address input with Google Places Autocomplete for accurate address capture.
- View list of matched travellers who have accepted the request.
- Select preferred traveller and confirm booking with payment.
- Real-time tracking of booking status (Confirmed → In Transit → Delivered).
- OTP generation and sharing for parcel pickup verification.
- Booking history with complete delivery records.

Traveller Module

The Traveller Module enables registered travellers to earn income by carrying parcels along their existing travel routes.

- Traveller Registration with KYC document submission (Aadhaar + PAN + Selfie).
- Route Creation with support for one-time and recurring weekly schedules.
- Multiple transport mode support: Private Vehicle, Bus, and Train.
- Automatic matching: system sends parcel requests matching the traveller's route.
- Accept or reject parcel requests; option to submit a counter-price offer.
- Manage active bookings and view pickup and delivery OTP flows.
- Earnings dashboard with per-booking payment records.

Admin Module

The Admin Module provides platform administrators with complete oversight and control capabilities.

- Admin Dashboard with platform-level statistics and summary metrics.
- KYC Management: review, approve, or reject traveller identity documents.
- User management: view, search, and manage registered users.
- Booking oversight: monitor all active and completed bookings.
- Route and parcel request monitoring across the platform.

Payment Module

The Payment Module manages all financial transactions on the platform.

- Pay Now: upfront payment via Razorpay before booking confirmation.
- Pay After Delivery: cash payment to the traveller upon successful delivery.
- Razorpay Order Creation and cryptographic payment signature verification.
- Automatic Booking Creation on confirmed payment.
- Refund Processing on cancellation or failure.

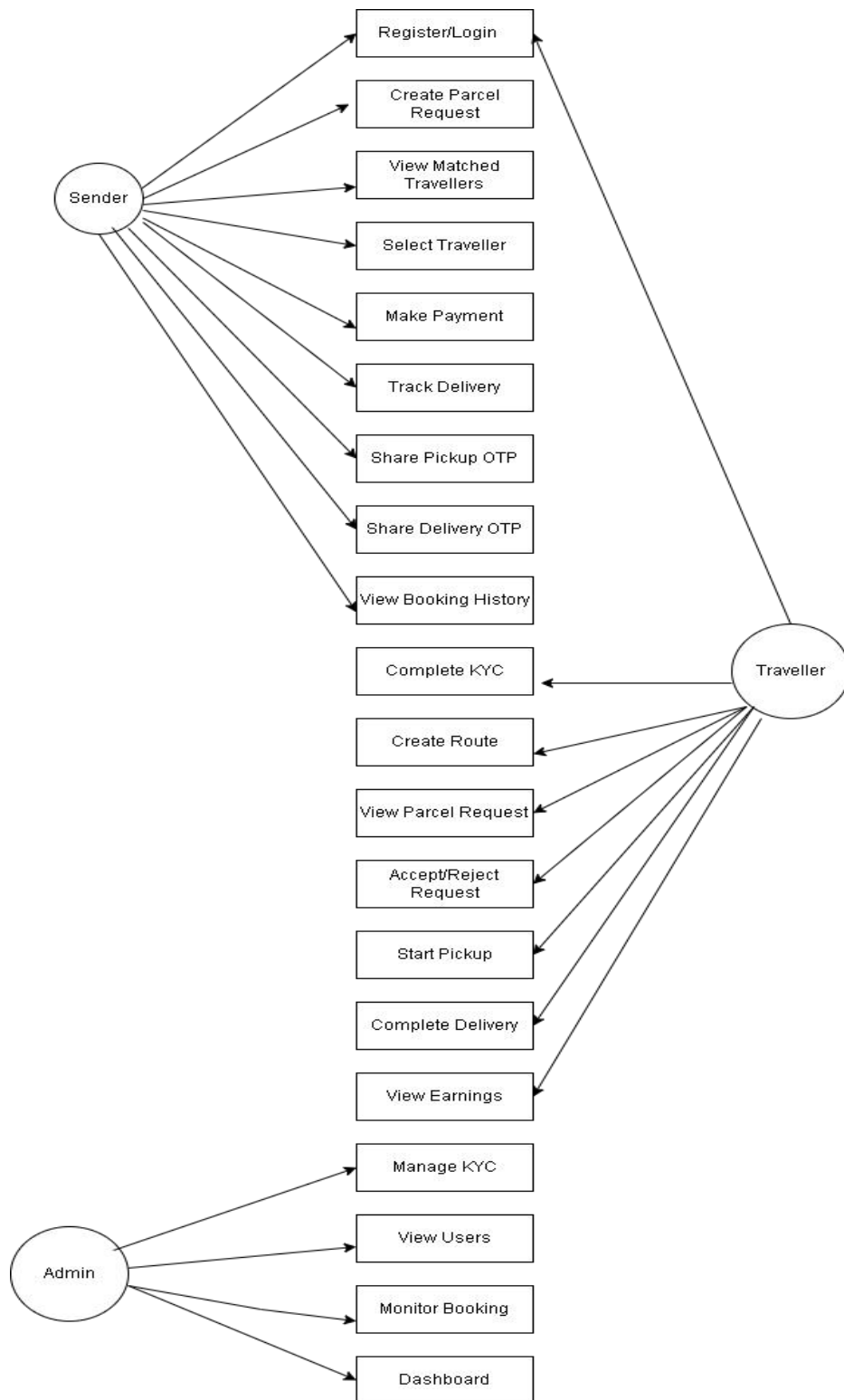
Notification Module

The Notification Module ensures all parties receive timely updates.

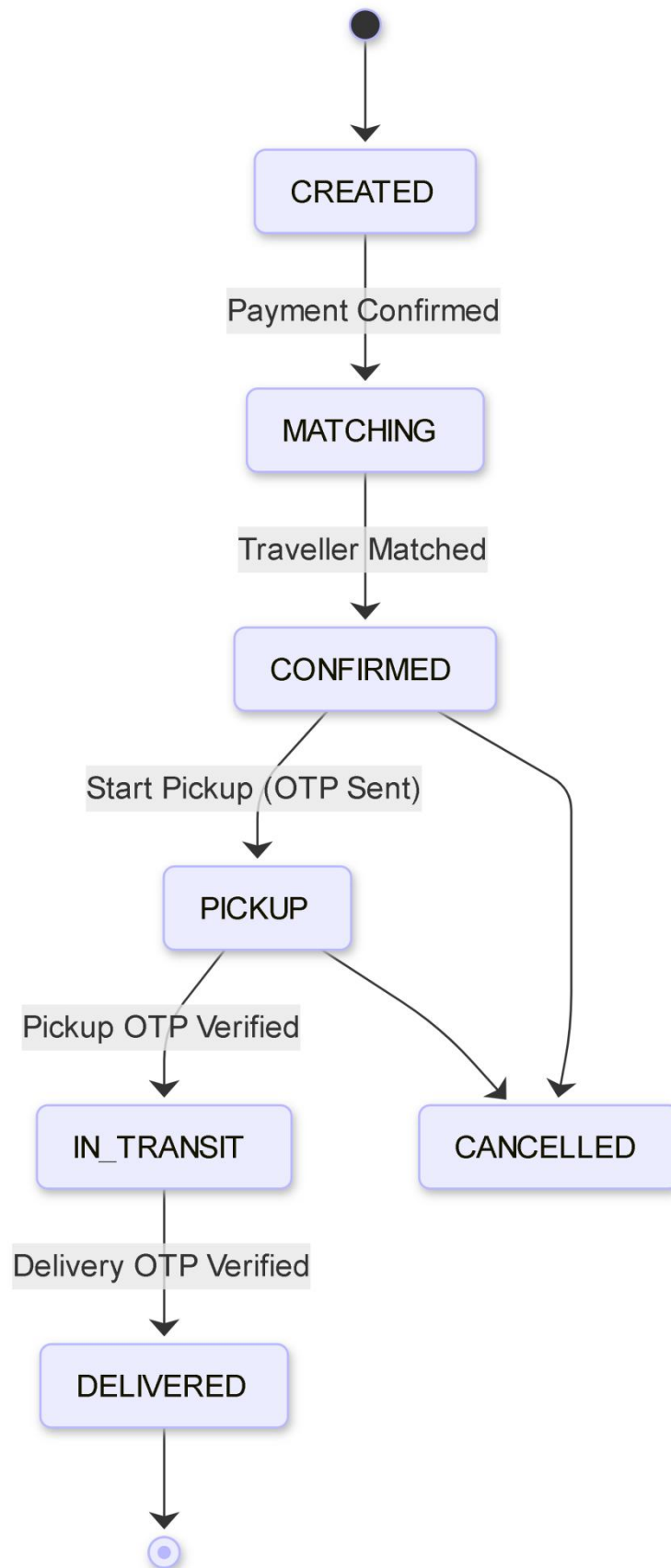
- In-app real-time notifications via Socket.io connections.
- Push notifications via Firebase Cloud Messaging for mobile and desktop browsers.
- SMS notifications via Twilio for OTP delivery and critical status updates.
- Notification history with read/unread tracking for each user.

CHAPTER 3: ANALYSIS AND DESIGN

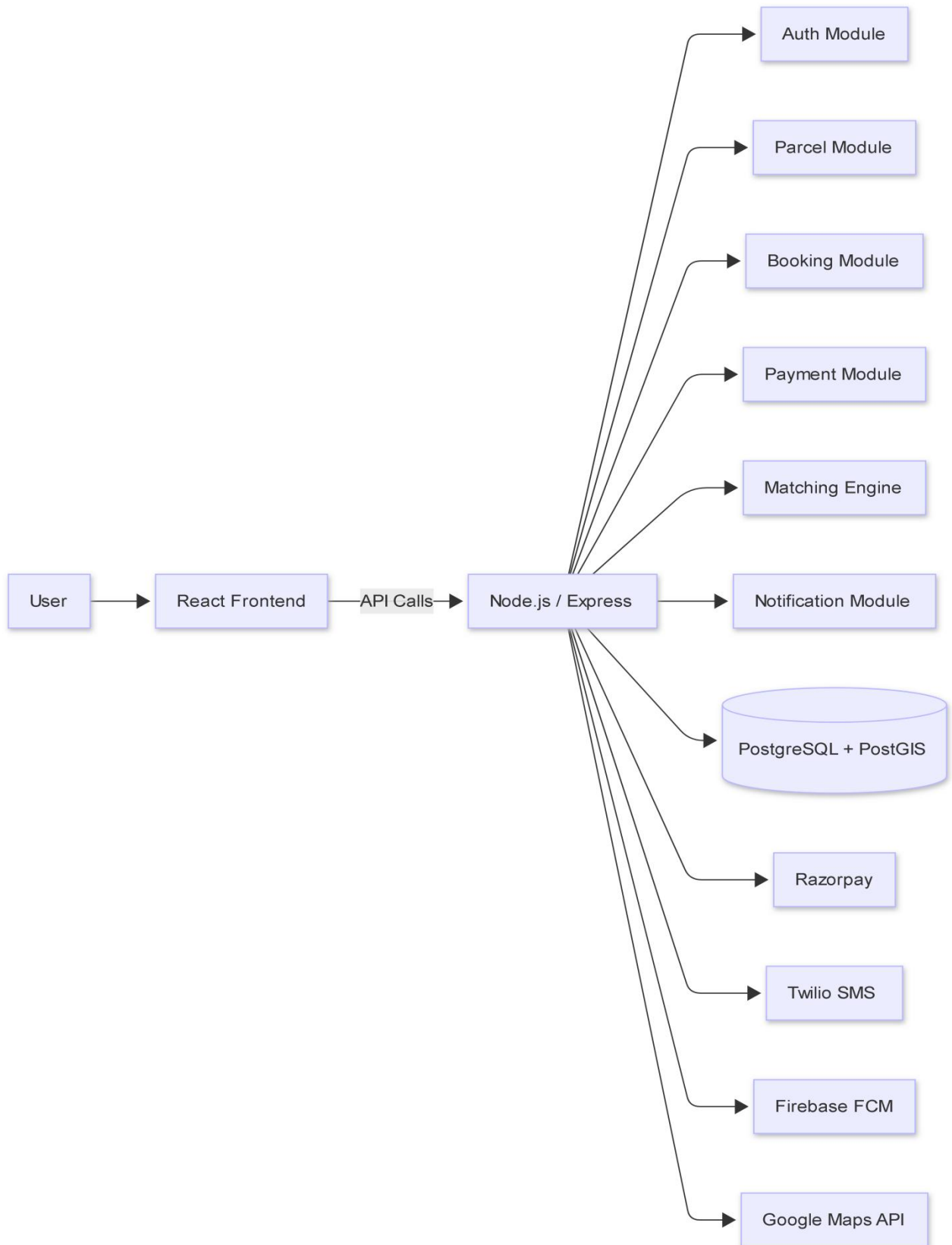
3.1 Use Case Diagram



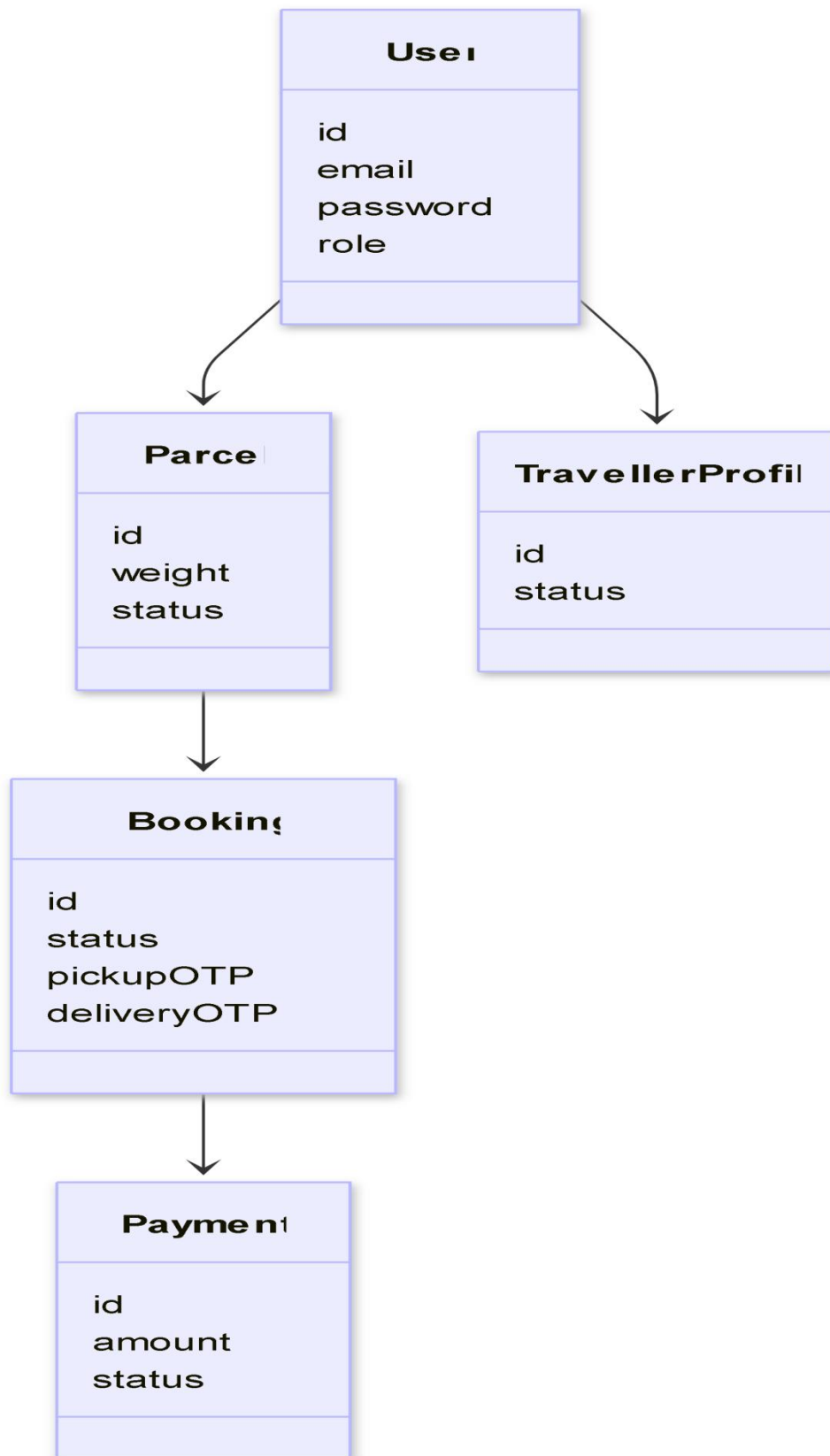
3.2 State Diagram



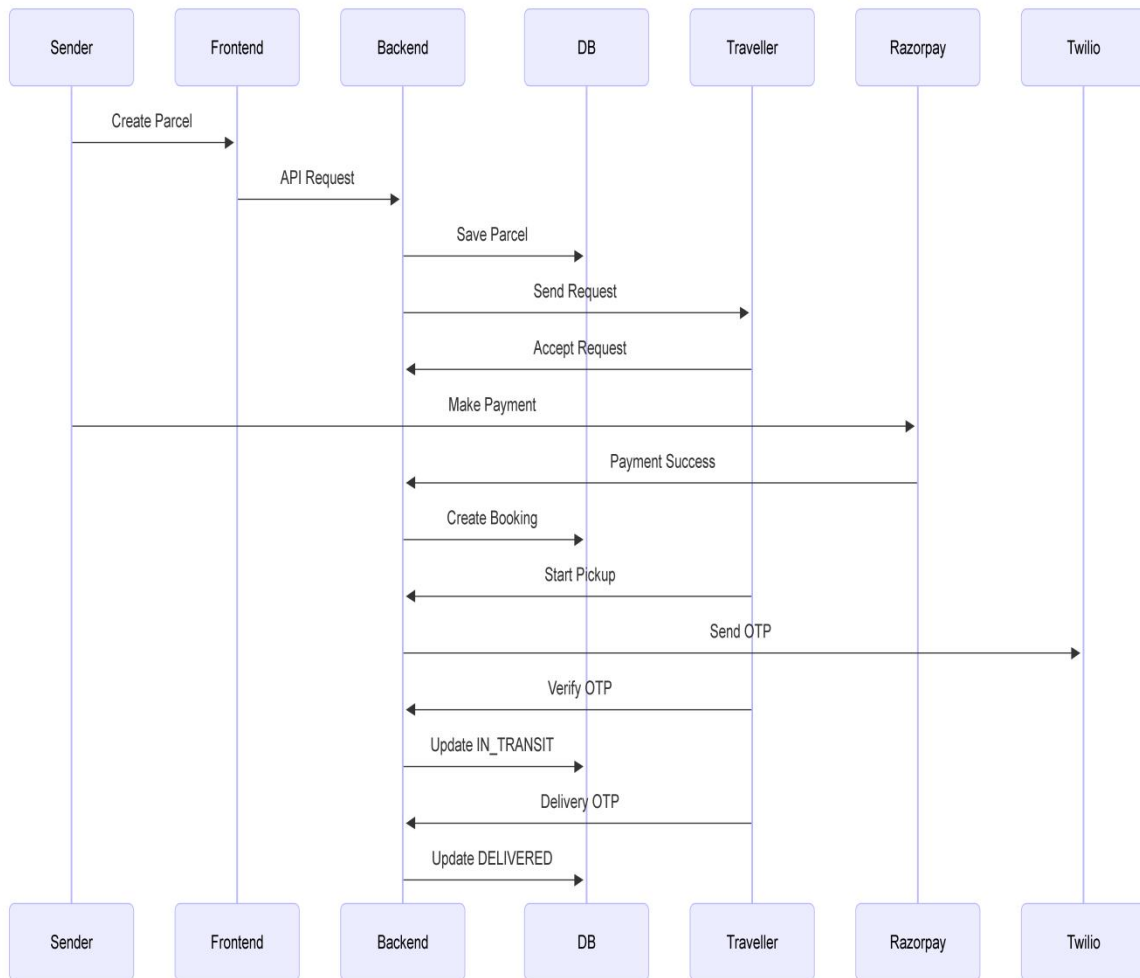
3.3 Component Diagram



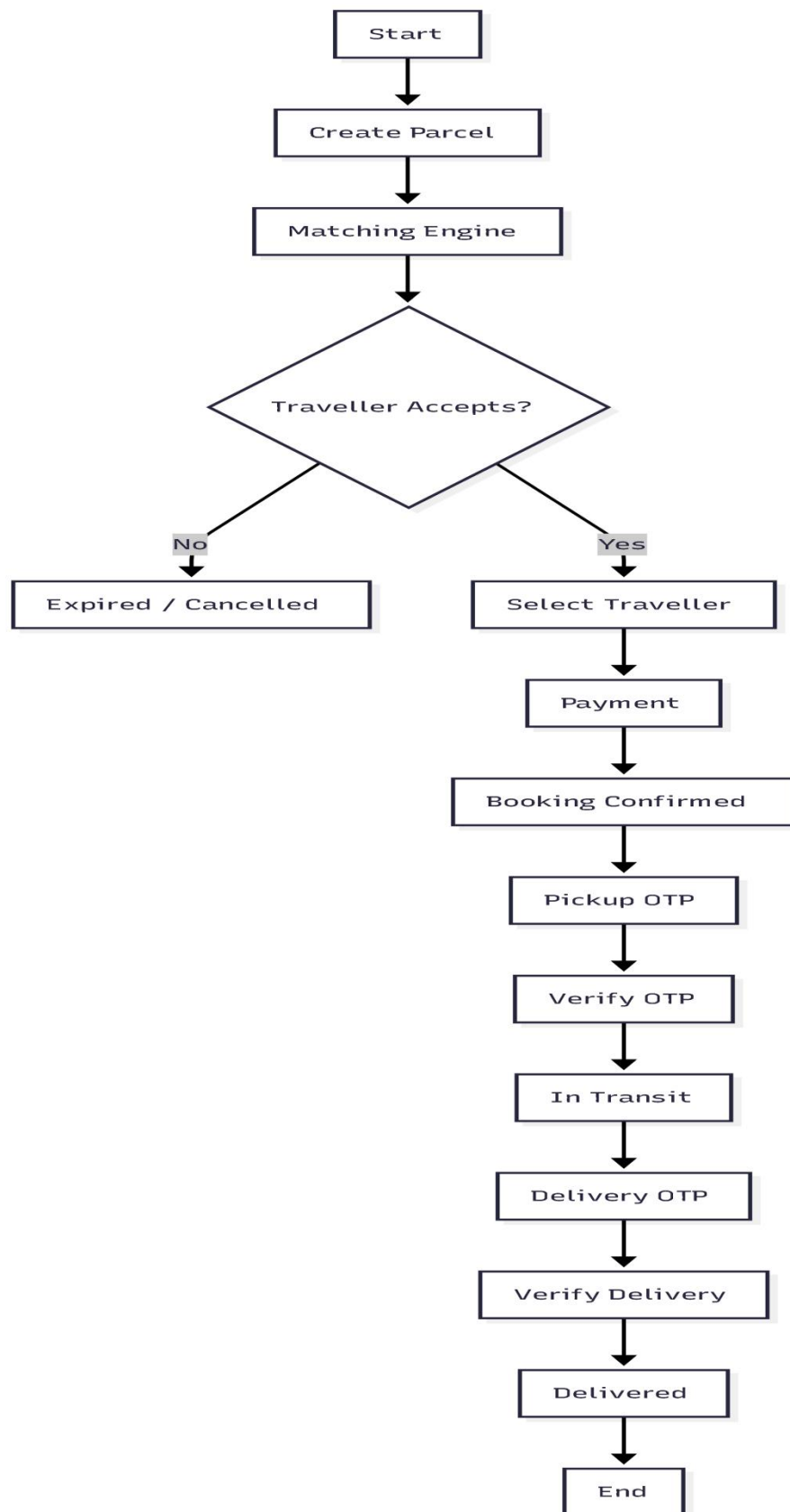
3.4 Class Diagram



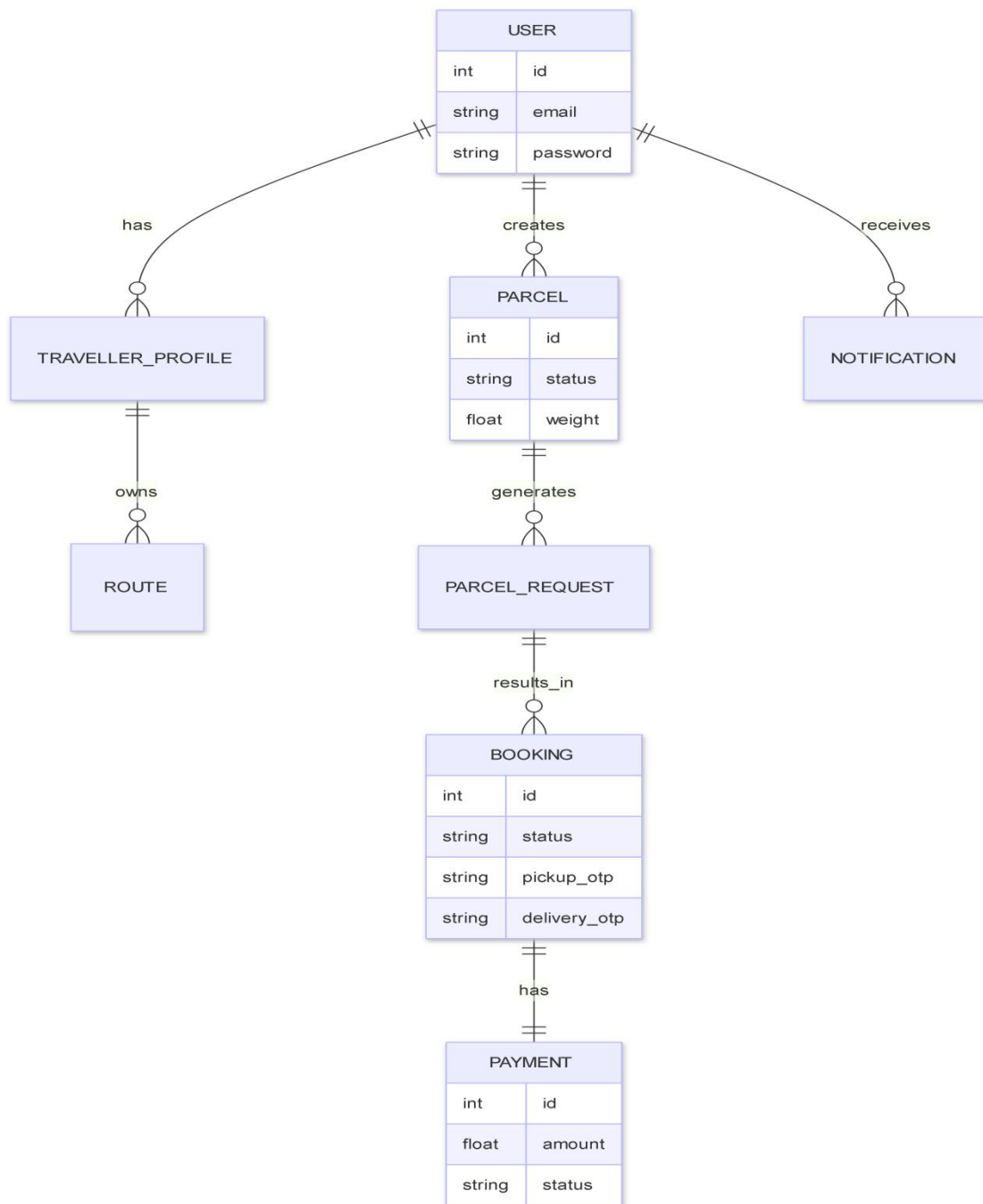
3.5 Sequence Diagram



3.6 Flow Chart



3.7 ER Diagram



3.8 Table Specifications

The following table describes the key database tables in the Book My Parcel system:

Table Name	Description	Key Fields
users	Central user accounts table storing authentication credentials and role assignments.	id, email, phone, password_hash, roles
user_profiles	Personal details of individual sender users.	user_id, full_name, profile_photo, date_of_birth
traveller_profiles	Traveller-specific profile data including capacity and preferences.	user_id, status, avg_rating, total_deliveries
traveller_routes	Routes registered by travellers including geometry and schedule.	id, user_id, origin_id, dest_id, transport_mode, is_recurring, route_geom
addresses	Enriched address records with geocoordinates and administrative hierarchy.	id, address_line, city, pincode, latitude, longitude, place_id, district
parcels	Core parcel delivery requests created by senders.	id, user_id, pickup_address_id, delivery_address_id, status, weight, price_quote
parcel_requests	Requests sent to matched travellers for acceptance.	id, parcel_id, traveller_id, route_id, status, detour_km, expires_at
parcel_acceptances	Records of travellers who accepted parcel requests.	id, parcel_request_id, traveller_id, acceptance_price
bookings	Confirmed delivery engagements between sender and traveller.	id, parcel_id, traveller_id, status, booking_ref, pickup_otp, delivery_otp
payments	Payment records linked to bookings.	id, booking_id, razorpay_order_id, amount, status, payment_mode
traveller_kyc	KYC document records for traveller identity verification.	id, user_id, aadhar_front_url, pan_front_url, selfie_url, status
notifications	Platform notifications for users and travellers.	id, user_id, role, title, message, type, is_read

CHAPTER 4: IMPLEMENTATION

The implementation chapter presents the key input screens and user interfaces of the Book My Parcel platform. All screens were developed using React.js with Redux Toolkit for state management and styled for a responsive, modern user experience.

4.1 Input Screens

Login and Registration Page

Book My Parcel

Start Sending Parcels Smarter

Join thousands of users sending parcels through travellers. Fast, secure and affordable.

One Account. Two Roles.

- Send parcels as an Individual
- Earn by delivering parcels
- Choose your role at login

Sign Up

Full Name *

Phone Number *

+91 Enter phone number

Email *

Enter your mail

City *

Enter city

State *

Enter state

Create Password *

Create a password

Confirm Password *

Confirm your password

Referral Code (optional)

Enter referral code

☐ I agree to the Terms & Conditions and Privacy Policy

Create Account

OR

Already have an account? [Login](#)

Sender Dashboard

Main dashboard showing active parcels, booking status, and notifications. Displays summary cards for pending requests, active deliveries, and completed bookings.

Parcel Creation – Step 1

Form for entering parcel details, pickup address, and delivery address with Google Places Autocomplete integration. Fields include parcel weight, description, and address inputs with map preview.




Send Parcel

1 Pickup & Package & Delivery Details — 2 Select Traveller — 3 Review

Pickup Details

 **Pickup Location**

Sender Name

Pickup Address

City / Village State Pincode

Country Phone number Alternate phone

Aadhaar number (optional)

Package Details

Select Package Size

Small
Documents, letters
0 - 1 kg

Medium
Books, clothes
1 - 5 kg

Large
Electronics, shoes
5 - 10 kg

Extra Large
Furniture parts
10 - 20 kg

Parcel Details

Weight (kg) Dimensions (cm) Length Width Height

Package Description (optional)

Parcel Photos

Parcel photo 1
Upload Parcel Photo
PNG, JPG up to 5MB

Parcel photo 2
Upload Parcel Photo
PNG, JPG up to 5MB

Parcel photo 3
Upload Parcel Photo
PNG, JPG up to 5MB

Parcel Value (₹)

Parcel Type

Additional Note

Delivery Details

Receiver Name

Delivery Address

City State Pincode

Country Phone Number Alternate Phone Number

Next: Select Partner

Order Summary



PICKUP DETAILS

Contact Name

Address

City / State

Phone

Alternate Phone

DELIVERY DETAILS

Contact Name

Address

City / State

Phone

Alternate Phone

PACKAGE DETAILS

Size

Weight

Value

Description

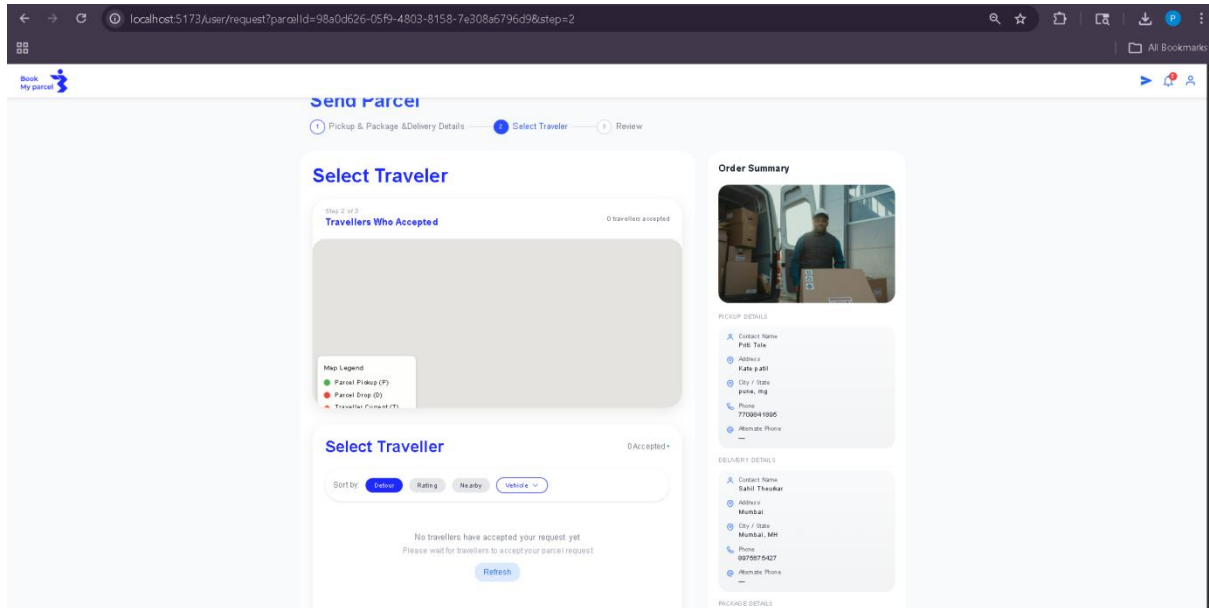
ESTIMATED PRICE

₹

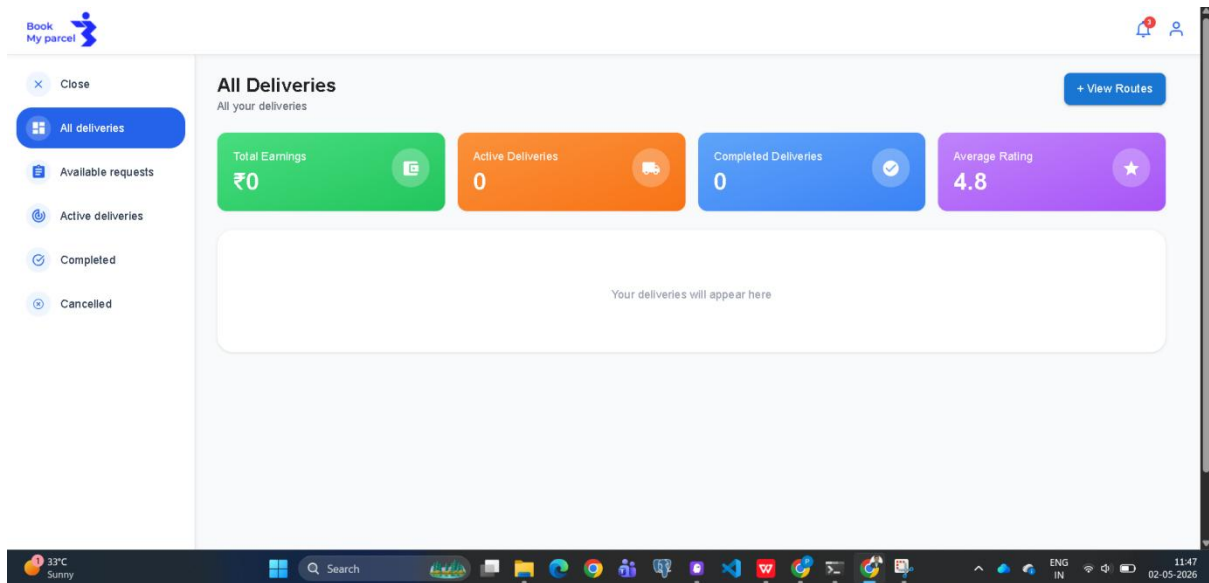
Includes all taxes & fees

Parcel Creation – Step 2 – Traveller Selection

List of matched travellers with profile, rating, price, and route details. Each card shows the traveller's verified status, transport mode, and estimated detour for the delivery.

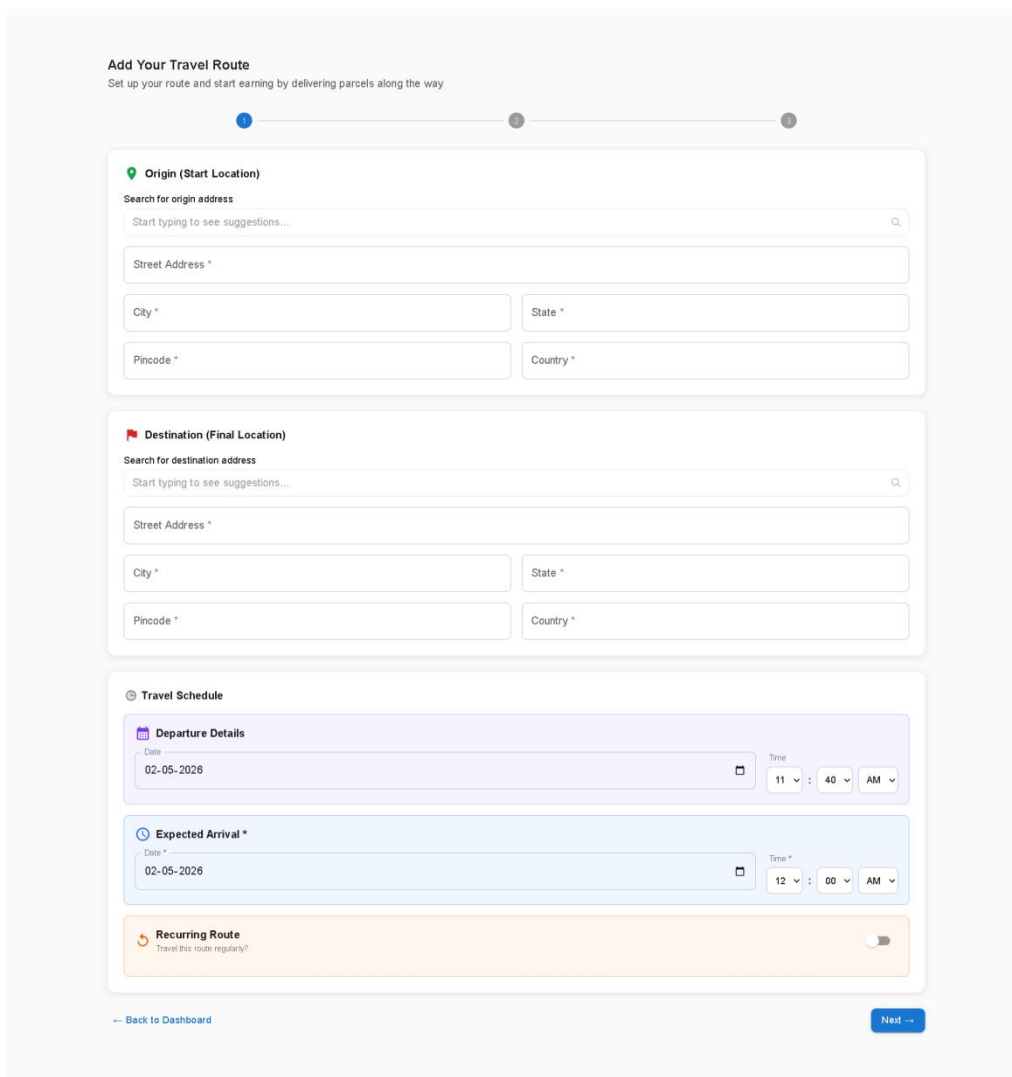


Traveller Dashboard



The screenshot shows a web application interface for a traveller dashboard. At the top left, there is a logo with the text "Book My parcel" and a blue icon. On the right, there are notification and user profile icons. A sidebar on the left contains a "Close" button and a list of filters: "All deliveries" (selected), "Available requests", "Active deliveries", "Completed", and "Cancelled". The main content area is titled "All Deliveries" with a subtitle "All your deliveries" and a "+ View Routes" button. Below this, there are four colored cards: "Total Earnings ₹0" (green), "Active Deliveries 0" (orange), "Completed Deliveries 0" (blue), and "Average Rating 4.8" (purple). A large white box in the center contains the text "Your deliveries will appear here". The bottom of the image shows a Windows taskbar with various application icons, system clock, and weather information (33°C Sunny).

Route Creation Form



The screenshot displays a "Route Creation Form" titled "Add Your Travel Route" with the subtitle "Set up your route and start earning by delivering parcels along the way". The form is divided into three main sections: "Origin (Start Location)", "Destination (Final Location)", and "Travel Schedule". Each section has a search bar and several input fields for address details (Street Address, City, State, Pincode, Country). The "Travel Schedule" section includes "Departure Details" and "Expected Arrival" with date and time pickers, and a "Recurring Route" toggle switch. Navigation buttons "Back to Dashboard" and "Next" are located at the bottom.

Add Your Travel Route
Set up your route and start earning by delivering parcels along the way

Origin (Start Location)
Search for origin address
Start typing to see suggestions...

Street Address *

City * State *

Pincode * Country *

Destination (Final Location)
Search for destination address
Start typing to see suggestions...

Street Address *

City * State *

Pincode * Country *

Travel Schedule

Departure Details
Date: 02-05-2026 Time: 11 : 40 AM

Expected Arrival *
Date: 02-05-2026 Time: 12 : 00 AM

Recurring Route
Travel this route regularly? ☐

[Back to Dashboard](#) [Next](#)

Parcel Request Detail

Book
My parcel

Parcel ID: BMP-001

Booking ID: IND091-001

Tracking ID: UBG-001

ROUTE SUMMARY

Pickup

Kate Patil Chowk, Nashik Phata Road, Sai Nagar Park, Pimpri-Chinchwad, Maharashtra, India, Pimpri-Chinchwad, Maharashtra, 411027

CONTACT SUMMARY

PHONE NUMBER

8975676542

ALTERNATE PHONE NUMBER

—

CONTACT NAME

Sahil Theurkar

PACKAGE INFORMATION

SIZE

medium

WEIGHT

1 kg

VALUE

₹200

SPEED

Standard

EST. DELIVERY

3-5 Days

DESC

Be Safe

DIMENSIONS

Length: 20 cm
Breadth: 20 cm
Width: 20 cm

NOTES

Safe

TYPE

document

PHOTOS

Photo 1

ROUTE SUMMARY

Delivery

Mumbai Central Railway Station Building, Mumbai Central, Mumbai, Maharashtra, India, Mumbai, Maharashtra, 400008

CONTACT SUMMARY

PHONE NUMBER

7709841895

ALTERNATE PHONE NUMBER

—

CONTACT NAME

Priti Tole

TRAVELER DETAILS

T

Traveller Sahil

★ 4.8 0 deliveries

VEHICLE

—

NUMBER

—

ROUTE

Pimpri-Chinchwad → Mumbai

EST. DELIVERY

Today

PRICE

₹137

PRICE BREAKDOWN

Base Price

₹137

Total

₹137

Decline

Booking Tracking View

Book
My parcel

Close

All Order

Active

Completed

Cancelled

069a25a9-63f3-4f24-b068-f1f99799c34f

Track

Select from your active deliveries

Estimated Arrival

~224 min

111.8 km remaining

Share Tracking Link

Map

Satellite

Mumbai

Pimpri-Chinchwad

Pune

Notification Page

Book
My parcel

Notifications

Stay updated on your delivery activities

Mark all as read (3)

All Notifications

Unread

Bookings

Offers

all

unread

booking

offer

Older

Parcel Picked Up

Your parcel has been picked up by the traveller. Booking ref: IND091-001

View

4/11/2026

Traveller Interested in Your Parcel

A traveller has expressed interest in delivering your parcel!

View

4/11/2026

Booking Cancelled

Your booking BK-2026-C6166D76 has been cancelled by the traveller.

View

4/11/2026

CHAPTER 5: TESTING

5.1 Test Procedures and Implementation

The testing strategy for Book My Parcel covers functional testing, integration testing, and end-to-end workflow testing. Manual testing was performed using Postman for API testing and browser-based testing for UI flows. All critical user journeys were verified against expected outcomes.

Test Types

- **Unit Testing:** Individual API endpoints tested for correct input validation, business logic, and response codes.
- **Integration Testing:** End-to-end flows tested including parcel creation, matching engine execution, acceptance, payment, and OTP verification.
- **Functional Testing:** All user-facing features tested against defined requirements for Sender, Traveller, and Admin roles.
- **Security Testing:** Authentication, authorisation, OTP lockout, and payment signature verification tested for correct behaviour.

Test Environment

- **API Testing Tool:** Postman with pre-configured environments for development and staging.
- **Payment Testing:** Razorpay sandbox environment with test card numbers and UPI IDs.
- **SMS Testing:** Twilio test credentials for OTP delivery simulation.
- **Database:** Separate test schema on PostgreSQL to avoid contaminating production data.

5.2 Test Cases

Test ID	Module	Description	Input	Expected Result	Status
TC-01	Auth	User registration with valid data	Name, email, phone, password	Account created, JWT token returned	PASS
TC-02	Auth	Login with correct credentials	Valid email + password	Login successful, token returned with role	PASS
TC-03	Auth	Login with incorrect password	Valid email + wrong password	HTTP 401 Unauthorized error	PASS
TC-04	Parcel	Create parcel with valid addresses	Valid pickup and delivery addresses	Parcel created, matching engine triggered	PASS
TC-	Parcel	Create parcel	Missing	HTTP 400	PASS

Test ID	Module	Description	Input	Expected Result	Status
05		without pickup address	pickup address	Validation error	
TC-06	Matching	Match traveller for intercity parcel	Mumbai-Pune parcel, Mumbai-Pune route	Matching travellers identified correctly	PASS
TC-07	Matching	No match for unrelated route	Mumbai-Pune parcel, Delhi-Agra route	Zero matches returned	PASS
TC-08	Route	Create one-time route with valid data	Valid origin, destination, date, transport mode	Route saved with geometry	PASS
TC-09	Route	Create recurring route with days	Weekday schedule with start/end dates	Recurring route saved with days array	PASS
TC-10	Accept	Traveller accepts parcel request	Valid request ID	Acceptance created, sender notified via Socket.io	PASS
TC-11	Booking	Booking created after payment	Valid payment signature from Razorpay	Booking confirmed, status updated	PASS
TC-12	OTP	Verify correct pickup OTP	Correct 4-digit OTP	Status changed to IN_TRANSIT	PASS
TC-13	OTP	Reject incorrect pickup OTP	Wrong 4-digit OTP	Error returned, attempt counter incremented	PASS
TC-14	OTP	OTP lockout after 5 failed attempts	5 consecutive wrong OTPs	Account locked for 1 hour	PASS
TC-15	OTP	Verify correct delivery OTP	Correct delivery OTP	Status changed to DELIVERED	PASS

Test ID	Module	Description	Input	Expected Result	Status
TC-16	KYC	Submit KYC documents	Valid image files for Aadhaar and PAN	KYC status updated to PENDING	PASS
TC-17	Admin	Admin approves KYC	Admin approval action	KYC status set to APPROVED, traveller activated	PASS
TC-18	Notif.	Real-time Socket.io notification	New parcel request created	Traveller receives real-time notification	PASS
TC-19	Payment	Razorpay payment with valid card	Test card details from Razorpay sandbox	Payment confirmed, booking created	PASS
TC-20	Security	Access protected route without token	No Authorization header	HTTP 401 Unauthorized response	PASS

CHAPTER 6: USER MANUAL

Getting Started

This manual provides step-by-step instructions for using the Book My Parcel platform for both Senders and Travellers.

For Senders – How to Send a Parcel

Step 1: Registration and Login

- Visit the Book My Parcel web application and click 'Register'.
- Select your role as 'Sender' and fill in your name, email, phone number, and password.
- Verify your email address and log in to access your dashboard.

Step 2: Create a Parcel Request

- From your dashboard, click 'Send a Parcel' to begin the 3-step creation wizard.
- Step 1 — Enter parcel details: weight, description, and pickup/delivery addresses using the autocomplete search.
- Step 2 — View matched travellers: browse available travellers, check their profile, rating, price, and route details.
- Step 3 — Payment: choose 'Pay Now' via Razorpay or 'Pay After Delivery' and confirm your booking.

Step 3: Track Your Parcel

- Monitor your booking status in real time from your dashboard (Confirmed → In Transit → Delivered).
- When the traveller is ready to pick up the parcel, you will receive a 4-digit OTP via SMS.
- Share this OTP with the traveller to confirm pickup. Repeat for delivery.

For Travellers – How to Carry Parcels

Step 1: Registration and KYC

- Register with the 'Traveller' role and submit your KYC documents: Aadhaar card, PAN card, and a selfie.
- Wait for admin KYC approval before you can receive parcel requests.

Step 2: Create a Route

- After KYC approval, go to 'My Routes' and click 'Add Route'.
- Enter your origin and destination cities, select your transport mode (Private Vehicle, Bus, or Train).
- Choose a one-time date or set up a recurring weekly schedule.

Step 3: Accept Parcel Requests

- Incoming parcel requests matching your route will appear in your dashboard with real-time notifications.
- Review the parcel details, detour distance, and offered price. Accept or reject the request.
- Optionally submit a counter-price offer for negotiation.

Step 4: Execute Pickup and Delivery

- Once a sender confirms booking with payment, proceed to the pickup location.
- Click 'Start Pickup' — an OTP SMS will be sent to the sender. Enter the OTP shared by the sender to verify pickup.

- Transport the parcel and click 'Start Delivery' at the destination. Enter the receiver's OTP to confirm delivery and mark the booking as completed.

For Administrators

- Log in with admin credentials to access the Admin Panel.
- Review pending KYC submissions from the KYC Management section.
- View uploaded documents (Aadhaar, PAN, Selfie) and click 'Approve' or 'Reject'.
- Monitor all bookings, users, routes, and parcel requests from the dashboard.

CHAPTER 7: DRAWBACKS AND LIMITATIONS

While Book My Parcel provides a comprehensive and innovative parcel delivery solution, certain limitations exist in the current implementation:

- **No Mobile Application:** The current version is a web application only. A dedicated native mobile app with offline support and live GPS tracking is not yet available.
- **Manual KYC Review:** The KYC approval process is entirely manual, requiring admin intervention. An automated document verification system has not been implemented.
- **No Live GPS Tracking:** While booking status updates are real-time, live location tracking of the traveller during the journey is not implemented in the current version.
- **No In-App Chat:** There is no direct messaging capability between senders and travellers within the platform. Communication must happen through external means.
- **No Wallet System:** Traveller earnings are tracked but there is no integrated wallet for withdrawals or balance management.
- **Internet Dependency:** The platform requires a consistent internet connection and does not support offline mode for any operations.
- **Limited Payment Methods:** Currently only Razorpay card/UPI payments and cash-on-delivery are supported. No wallet or net banking integrations beyond Razorpay's built-in support.
- **No AI-Based Fraud Detection:** The system lacks automated fraud detection for suspicious booking patterns or fake KYC submissions.

CHAPTER 8: PROPOSED ENHANCEMENTS

The following enhancements are planned for future versions of the Book My Parcel platform:

- **Native Mobile Application:** Develop iOS and Android applications using React Native to reach a wider audience with features like live GPS tracking, push notifications, and offline caching.
- **Live Location Tracking:** Integrate real-time GPS sharing during the delivery journey, allowing senders to track parcel location on a live map.
- **AI-Powered Matching:** Implement machine learning models that improve match scoring based on traveller historical performance, delivery success rates, and route reliability.
- **In-App Chat System:** Add real-time messaging between senders and travellers within the platform, reducing reliance on phone calls.
- **Automated KYC Verification:** Integrate third-party identity verification APIs (e.g., DigiLocker, Aadhaar eKYC) for instant automated document validation.
- **Traveller Wallet:** Implement a digital wallet for travellers to receive, hold, and withdraw earnings directly through the platform, with UPI and bank transfer support.
- **Dynamic Pricing Engine:** Introduce demand-based and route-based dynamic pricing to optimise earnings for travellers and offer competitive rates for senders.
- **Multi-Parcel Optimization:** Develop route optimization for travellers who wish to carry multiple parcels simultaneously on a single trip.
- **Rating and Review System:** Allow senders to rate travellers and vice versa after each delivery, building a community reputation system.
- **Analytics Dashboard:** Provide travellers with detailed earnings analytics and senders with delivery performance metrics.
- **Enterprise API:** Develop a B2B API for small businesses and e-commerce platforms to integrate Book My Parcel as a delivery option at checkout.
- **International Expansion:** Extend the platform architecture to support cross-border deliveries and international traveller routes.

CHAPTER 9: CONCLUSION

Book My Parcel represents an innovative and practical application of technology to solve a real-world logistics challenge faced by millions of people across India. By creating a collaborative economy platform that connects parcel senders with everyday travellers, the system disrupts the traditional courier model and offers a cost-effective, community-driven alternative.

The project successfully demonstrates the implementation of complex full-stack web development concepts, including geospatial database queries with PostGIS, multi-stage algorithmic matching, real-time communication with Socket.io, secure payment processing, digital identity verification workflows, and integration with multiple third-party APIs including Google Maps Platform, Razorpay, Twilio, and Firebase.

The 5-stage matching engine is the technical centerpiece of the platform, combining geographic proximity, temporal alignment, capacity constraints, and detour analysis to identify the most suitable travellers for each parcel request. The OTP-based handover verification system provides a secure and verifiable chain of custody for every parcel handled on the platform.

While certain features such as a native mobile application, live GPS tracking, and automated KYC verification remain as future enhancements, the current implementation provides a complete, end-to-end working platform that is ready for real-world testing and deployment.

In conclusion, Book My Parcel fulfills all stated objectives and demonstrates a strong understanding of modern software development practices, system architecture, and user-centric design. The platform has the potential to create meaningful social and economic impact by empowering travellers and making parcel delivery more accessible and affordable for everyone.

CHAPTER 10: ANNEXURE

10.1 Technology Used

Frontend Technologies

- React.js (v18.x): Component-based JavaScript library for building the user interface. Enables a dynamic, single-page application experience with efficient DOM updates.
- Redux Toolkit (v1.9.x): State management library for managing global application state including authentication tokens, parcel data, and real-time notifications.
- Axios: HTTP client for making REST API requests from the frontend to the Node.js backend.
- Socket.io Client: Socket.io client library for receiving real-time booking status updates and notifications.
- Vite (v5.x): Modern build tool and development server offering fast hot module replacement during development.
- Google Maps JavaScript API: Integrated for address autocomplete, map display, and route visualisation.
- Razorpay Checkout: Client-side payment integration for secure card and UPI payment processing.

Backend Technologies

- Node.js (v18.x LTS): JavaScript runtime environment for executing the backend server-side code.
- Express.js (v4.x): Lightweight web framework for building RESTful API endpoints and middleware chains.
- Socket.io (v4.x): Socket.io library enabling real-time bidirectional communication between server and connected clients.
- Sequelize (v6.x): ORM (Object-Relational Mapper) for defining database models and executing SQL queries with JavaScript.
- JSON Web Tokens (JWT): Stateless authentication mechanism for securing API endpoints based on user role.
- bcrypt: Password hashing library for secure storage of user credentials.
- Multer: Middleware for handling multipart/form-data file uploads (KYC documents and profile photos).

Database Technologies

- PostgreSQL (v15.x): Robust open-source relational database system used for storing all platform data.
- PostGIS (v3.x): Spatial extension for PostgreSQL enabling geographic queries, buffer-based proximity matching, and route geometry storage.

External Services and APIs

- Google Maps Platform: Address Validation API, Geocoding API, Places API (New), and Routes API for address enrichment, geocoding, and route calculation.
- Razorpay Payment Gateway: Payment order creation, signature verification, refund processing, and webhook event handling.
- Twilio SMS API: OTP SMS delivery to senders and receivers during pickup and delivery verification.
- Firebase Cloud Messaging (FCM): Push notification delivery to browser clients for

real-time booking and delivery alerts.

Deployment Infrastructure

- Vercel: Frontend deployment platform offering automatic deployments from GitHub, global CDN, and serverless edge functions.
- Render: Backend deployment platform for Node.js services with automatic SSL, health checks, and environment variable management.

CHAPTER 11: BIBLIOGRAPHY

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- PostGIS Official Documentation — <https://postgis.net/documentation>
- Sequelize ORM Documentation — <https://sequelize.org/docs>
- Socket.io Documentation — <https://socket.io/docs>
- Redux Toolkit Documentation — <https://redux-toolkit.js.org>

External Service Documentation

- Google Maps Platform Documentation — <https://developers.google.com/maps/documentation>
- Google Geocoding API — <https://developers.google.com/maps/documentation/geocoding>
- Google Places API (New) — <https://developers.google.com/maps/documentation/places>
- Google Routes API — <https://developers.google.com/maps/documentation/routes>
- Razorpay Payment Gateway Documentation — <https://razorpay.com/docs>
- Twilio SMS API Documentation — <https://www.twilio.com/docs/sms>
- Firebase Cloud Messaging Documentation — <https://firebase.google.com/docs/cloud-messaging>

Academic and Learning Resources

- Vite Build Tool Documentation — <https://vitejs.dev/guide>
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